

Module 1: Getting Started

Exercise 1: Configuring a Storage VM

In this exercise, you verify that the required NFS licenses are installed, verify that an aggregate is created, and create two storage VMs (storage virtual machines, also known as SVMs).

Objectives

This exercise focuses on enabling you to do the following:

- Examine NFS licensing on the NetApp ONTAP cluster
- Verify that an aggregate is created
- Create a storage VM for NFS

Task 1: Examine NFS Licensing on the ONTAP Cluster

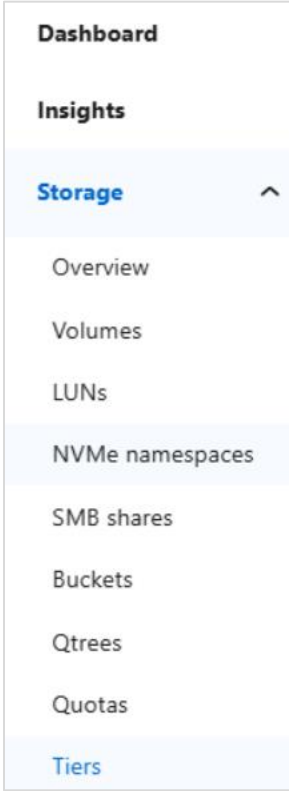
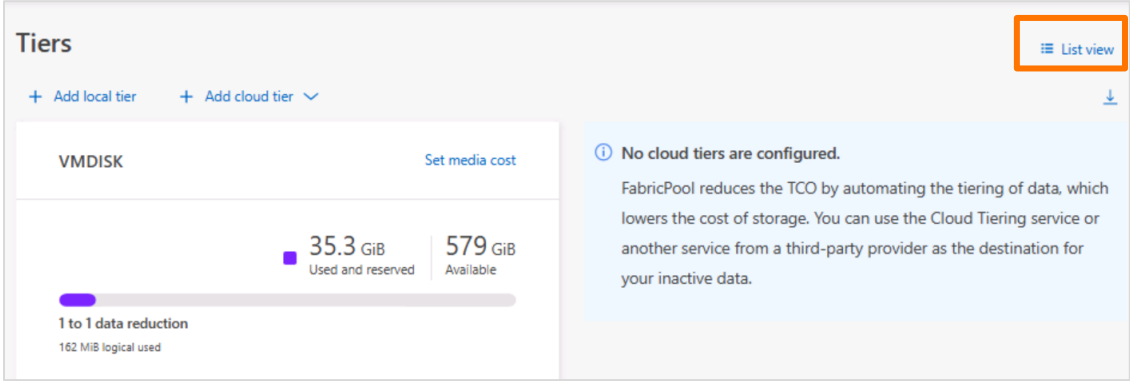
Many advanced features of ONTAP software require licenses. For later exercises, you use NFS, which is a licensed feature of ONTAP software. You can check licenses in NetApp ONTAP System Manager or the ONTAP CLI. In this task, you use the CLI to verify that the necessary licenses are installed.

Step	Action
1-1	In the cluster management CLI, run the following command: <code>license show</code>
1-2	Verify that (at least) the following required license codes—for the licenses that enable the features of ONTAP software and that are required for later exercises—are installed: • NFS • CIFS • S3
1-3	 If any license is not installed, inform your instructor.

Task 2: Verify that an Aggregate or Local Tier is Created

In this task, you log in to the ONTAP System Manager UI and review the aggregate or local tier information.

Step	Action
2-1	Go to the NetApp ONTAP System Manager window for cluster1.

Step	Action
2-2	In the navigation pane on the left, select Storage > Tiers. 
2-3	Click List view at the top right to change the view layout. 

Step

Action

2-4

Observe that the local tiers are already created.

Tiers

Local tiers

Cloud tiers

Storage pools

+ Add

Search

Download

Grid

Menu

Name	Status	Type	Capacity	Use...	Log...	Use...	Mirror
n2_ssd_1	Online	SSD	48.4 MiB used and reserved 76.7 GiB available	0 Byte	216 Kil	0%	Disabled
n1_ssd_1	Online	SSD	48.1 MiB used and reserved 76.7 GiB available	0 Byte	268 Kil	0%	Disabled

2-5

Click local tier **n1_ssd_1** and verify the following details on the **Overview** tab:

- Disk type
- RAID type
- Available and used capacity

← Back to Tiers

n1_ssd_1

Overview

Disk information

Volumes

Status

Online

Node

cluster1-01

Type

SSD (5 disks)

RAID

RAID-DP

Encryption status

Disabled

FabricPool

Disabled

Flash Pool

Disabled

SnapLock

Disabled

Mirror

Disabled

Capacity

40.8 MiB

Used and reserved

76.7 GiB

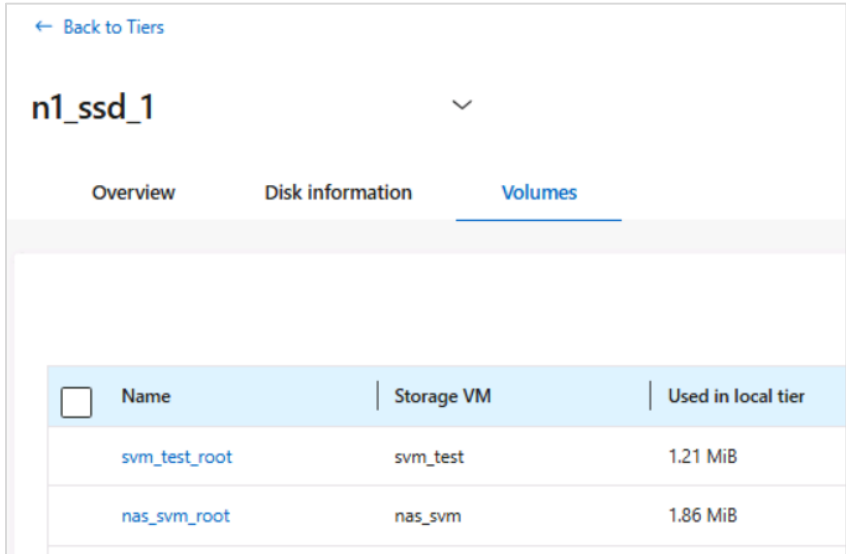
Available

40 MiB

Committed

1 to 1 data reduction

424 KiB logical used

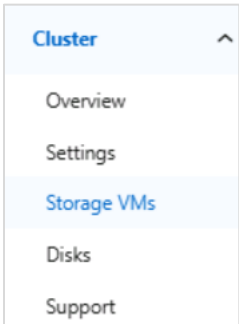
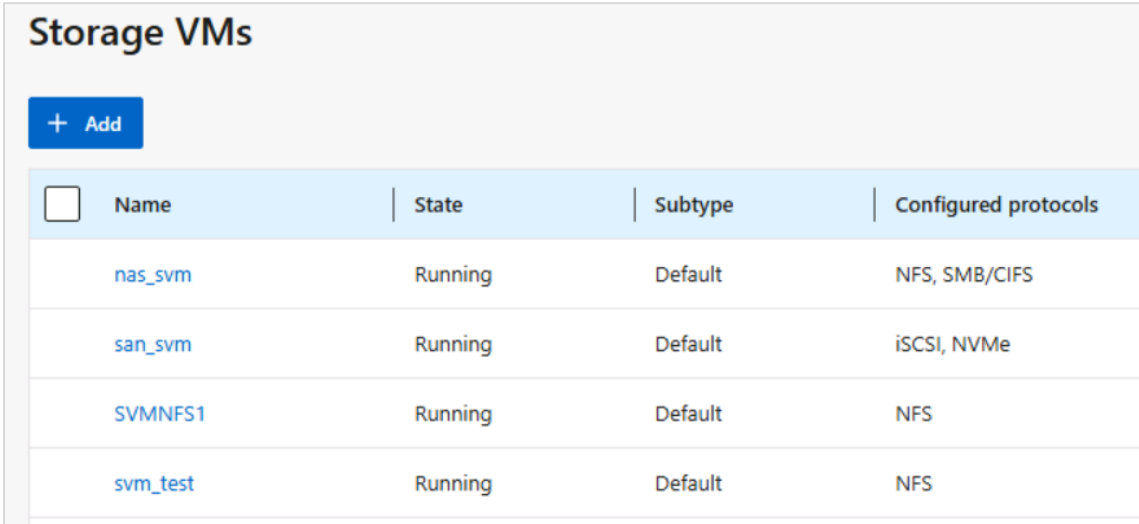
Step	Action												
2-6	Select the Volumes tab and check for any volumes that are assigned to the aggregate.  The screenshot shows the 'Volumes' tab for the aggregate 'n1_ssd_1'. It displays a table with the following data: <table><thead><tr><th>☐</th><th>Name</th><th>Storage VM</th><th>Used in local tier</th></tr></thead><tbody><tr><td>☐</td><td>svm_test_root</td><td>svm_test</td><td>1.21 MiB</td></tr><tr><td>☐</td><td>nas_svm_root</td><td>nas_svm</td><td>1.86 MiB</td></tr></tbody></table>	☐	Name	Storage VM	Used in local tier	☐	svm_test_root	svm_test	1.21 MiB	☐	nas_svm_root	nas_svm	1.86 MiB
☐	Name	Storage VM	Used in local tier										
☐	svm_test_root	svm_test	1.21 MiB										
☐	nas_svm_root	nas_svm	1.86 MiB										
2-7	Similarly, verify the other local tiers.												

Task 3: Create Storage VMs

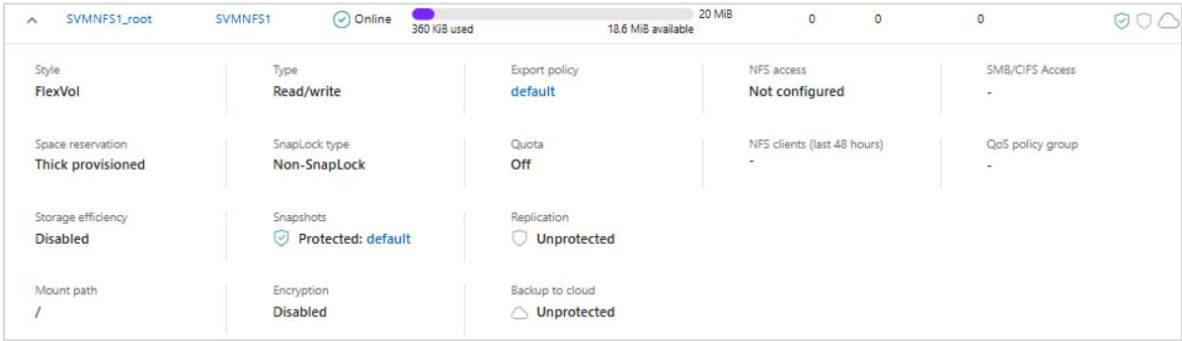
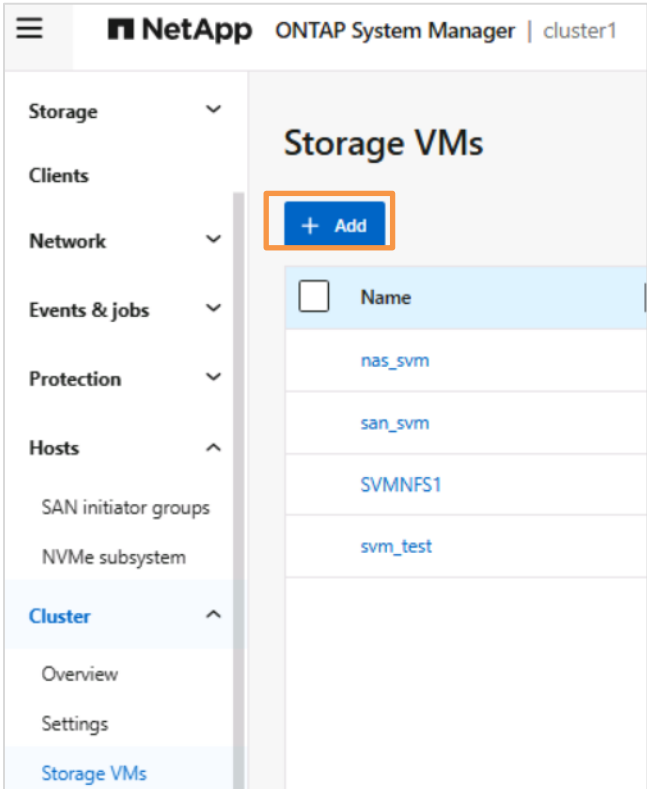
In this task, you create two storage VMs, one by using the cluster management CLI and another by using System Manager.

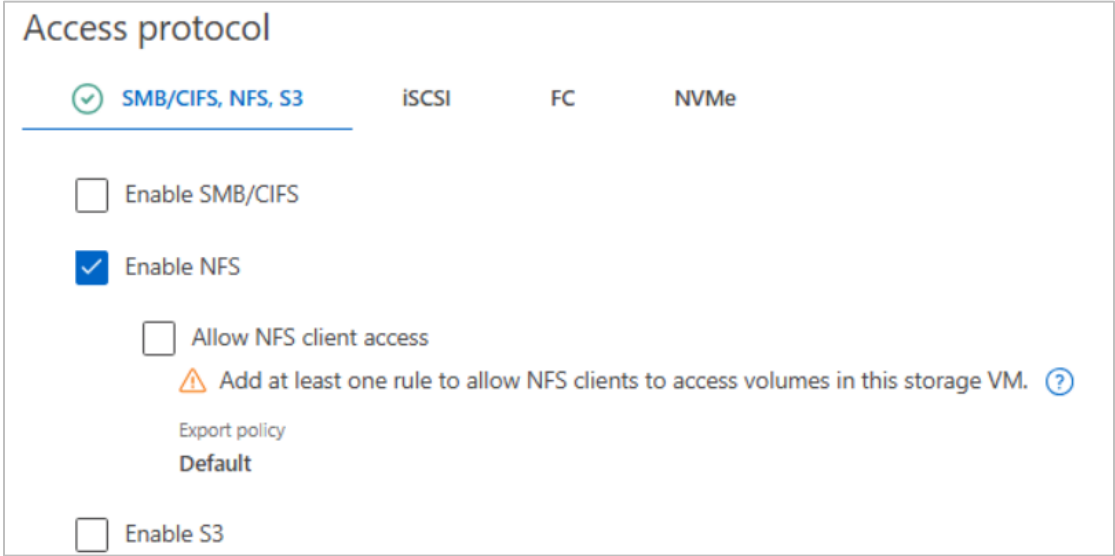
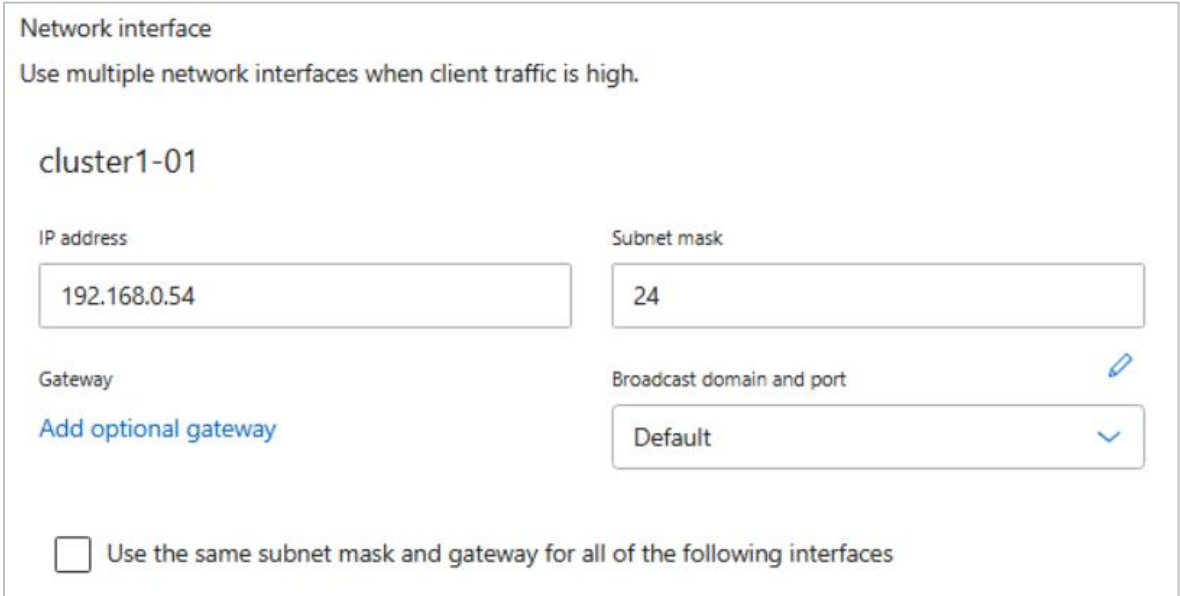
Step	Action
3-1	In the cluster management CLI, run the following command to create an SVM: <pre>vserver create -vserver SVMNFS1 -rootvolume SVMNFS1_root -aggregate n1_ssd_1 -rootvolume-security-style unix -language C.UTF_8</pre> Sample output: <pre>cluster1::> vserver create -vserver SVMNFS1 -rootvolume SVMNFS1_root - aggregate n1_ssd_1 -rootvolume-security-style unix -language C.UTF_8 [Job 741] Job succeeded: Vserver creation completed.</pre>
3-2	Run the following command to verify the SVM configuration: <pre>vserver show -vserver SVMNFS1</pre>

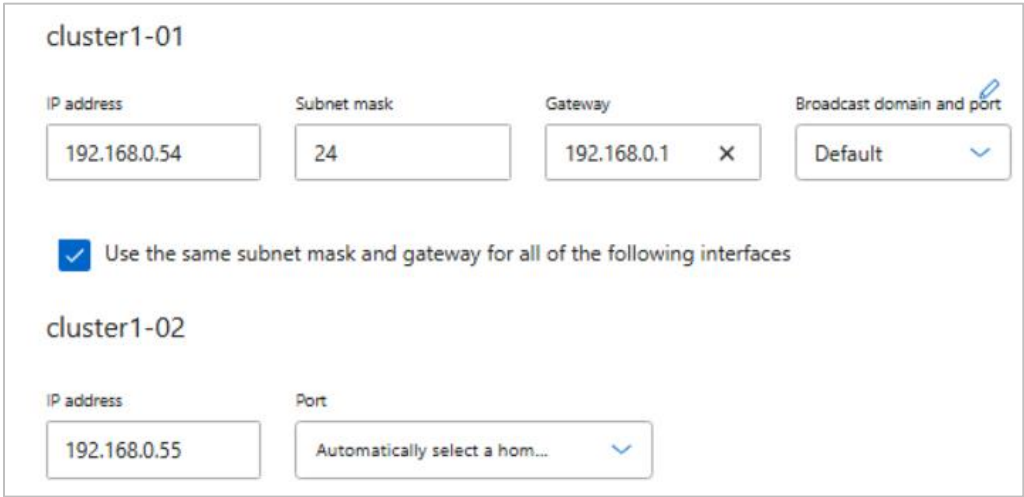
Step	Action
3-3	Verify the following parameters from the output: • Vserver Admin state is running • Allowed Protocols include all the protocols (NFS, CIFS, FCP, iSCSI, NDMP, S3) • Disallowed Protocols include NVMe <pre> cluster1::> vserver show -vserver SVMNFS1 Vserver: SVMNFS1 Vserver Type: data Vserver Subtype: default Vserver UUID: a315c60d-ba3f-11f0-8c88-005056ab3872 Root Volume: SVMNFS1_root Aggregate: n1_ssd_1 NIS Domain: - Root Volume Security Style: unix LDAP Client: - Default Volume Language Code: C.UTF-8 Snapshot Policy: default Data Services: data-cifs, data-flexcache, data-iscsi, data-nfs, data-nvme-tcp Comment: Quota Policy: default List of Aggregates Assigned: - Limit on Maximum Number of Volumes allowed: unlimited Vserver Admin State: running Press <space> to page down, <return> for next line, or 'q' to quit... </pre>
3-4	 If the output displays <code>Vserver Admin state</code> in initializing state, it indicates that some intermediate operation (such as root volume creation) failed and you might need to delete the SVM and re-create it.

Step	Action
3-5	Run the following command to create an NFS server on the SVMNFS1 storage VM and enable NFSv3 and NFSv4.1: <pre>cluster1::> vserver nfs create -vserver SVMNFS1 -v3 enabled -v4.1 enabled -v4-id-domain example.com</pre>
3-6	Run the following command to verify the NFS server status: <pre>cluster1::> vserver nfs status -vserver SVMNFS1</pre> The NFS server is running on Vserver "SVMNFS1".
3-7	Run the following command to verify the NFS server configuration: <pre>cluster1::> vserver nfs show -vserver SVMNFS1</pre>
3-8	From the output, verify that NFSv3 and NFSv4.1 are enabled.
3-9	From the System Manager menu, select Clusters > Storage VMs. 
3-10	Verify that the newly created storage VM is shown. 
3-11	Click SVMNFS1 .

Step	Action																												
3-12	Select the Settings tab, review the details, and answer the following questions: - Which protocols are configured? - Which authentication services are configured? - Which policies are configured? ← Back to storage VMs SVMNFS1 Overview Settings Replication File system Protocols NFS ⚠️ A network interface wasn't created, so the client can't access NFS service. Status ✔️ Enabled Version support 3, 4, 4.1, 4.2 🔗 ✎ SMB/CIFS 🔗 ⚙️ Status ❌ Not configured																												
3-13	Select Storage > Volumes .																												
3-14	Click the down arrow that corresponds to the SVMNFS1_root volume. Volumes Export volumes as file systems for NFS and SMB clients. More + Add ⋮ More <table><tr><th>☐</th><th>Name</th><th>Storage VM</th><th>Status</th></tr><tr><td>▼</td><td>nas_vol1</td><td>nas_svm</td><td>✔️ Online</td></tr><tr><td>▼</td><td>nas_vol2</td><td>nas_svm</td><td>✔️ Online</td></tr><tr><td>▼</td><td>san_svm_root</td><td>san_svm</td><td>✔️ Online</td></tr><tr><td>▼</td><td>san_vol1</td><td>san_svm</td><td>✔️ Online</td></tr><tr><td>▼</td><td>san_vol2</td><td>san_svm</td><td>✔️ Online</td></tr><tr><td>▼</td><td>SVMNFS1_root</td><td>SVMNFS1</td><td>✔️ Online</td></tr></table>	☐	Name	Storage VM	Status	▼	nas_vol1	nas_svm	✔️ Online	▼	nas_vol2	nas_svm	✔️ Online	▼	san_svm_root	san_svm	✔️ Online	▼	san_vol1	san_svm	✔️ Online	▼	san_vol2	san_svm	✔️ Online	▼	SVMNFS1_root	SVMNFS1	✔️ Online
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▼	SVMNFS1_root	SVMNFS1	✔️ Online																										

Step	Action
3-15	Review the mount path and NFS access. 
3-16	Select Clusters > Storage VMs .
3-17	Click + Add. 
3-18	In the Add storage VM dialog box, enter SVM1 in the Storage VM name field. 

Step	Action
3-19	Select the Enable NFS checkbox. 
3-20	In the Network interface section, specify the following settings for cluster1-01: - IP address: 192.168.0.54 - Subnet mask: 24 
3-21	Click Add optional gateway .
3-22	Select the Use the same subnet mask and gateway for all of the following interfaces checkbox.

Step	Action
3-23	In the Network Interface section, enter IP address 192.168.0.55 for cluster1-02. 
3-24	Click Save .
3-25	Review the information on the Storage VMs page.

End of exercise